



Mandi Mitra

An offline-first digital marketplace for farmers in low-connectivity regions

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Farmer Action

Local Database

Sync Queue

FastAPI

MongoDB Cloud

A Abstract

Mandi Mitra is a web-based marketplace that lets farmers list, price and trade crops even where internet access is unreliable. Actions and inventory updates are saved on the device first and synchronised to a central **MongoDB** database through a **FastAPI** backend once the network returns.

The design focuses on data availability, consistency, transaction management and reliable storage.

P Problem Statement & Objectives

Problem: Farmers in rural and semi-urban areas face unstable networks, low speeds and outages. Online-first marketplaces treat the server as the only source of truth, so an update such as *Rice, 500 kg, ₹50/kg* simply fails when the network drops.

- Objective 1:** Let farmers list products, prices and quantities offline.
- Objective 2:** Queue changes locally and sync them automatically when online.
- Objective 3:** Keep a consistent central MongoDB store for farmers and buyers.
- Scope:** Farmer and buyer modules: auth, products, search and messaging.

E Existing vs Proposed System

Feature	Existing	Proposed
Connectivity	Needs constant internet	Works offline
Data storage	Server only	Local first, then cloud
Updates	Fail on network loss	Saved and queued
Sync	Manual retry	Automatic on reconnect
Gap	Lost entries, low trust	Reliable, farmer-friendly

F Feasibility Study

- Technical:** Python, FastAPI, MongoDB and HTML/CSS/JS are open-source and well documented.
- Economic:** Free tools and free-tier cloud hosting keep the cost minimal.
- Operational:** Simple phone + PIN login and a clean, responsive interface.
- Schedule:** Staged build fits within the semester timeline.

R Results & Testing

- Product add, view, edit and delete working.
- Phone/PIN auth with hashed PINs.
- Pydantic validation, search and filter.

C Conclusion

- A farmer-first marketplace with a FastAPI + MongoDB core.
- Offline-first design keeps trading possible without a network.

N Future Scope

- Local DB and automatic sync engine.
- Payments, alerts and price-trend insights.
- Cloud deployment.

T Tools & Technologies

FRONT-END

HTML

CSS

JavaScript

BACK-END

Python

FastAPI

Pydantic

Uvicorn

DATABASE & TOOLS

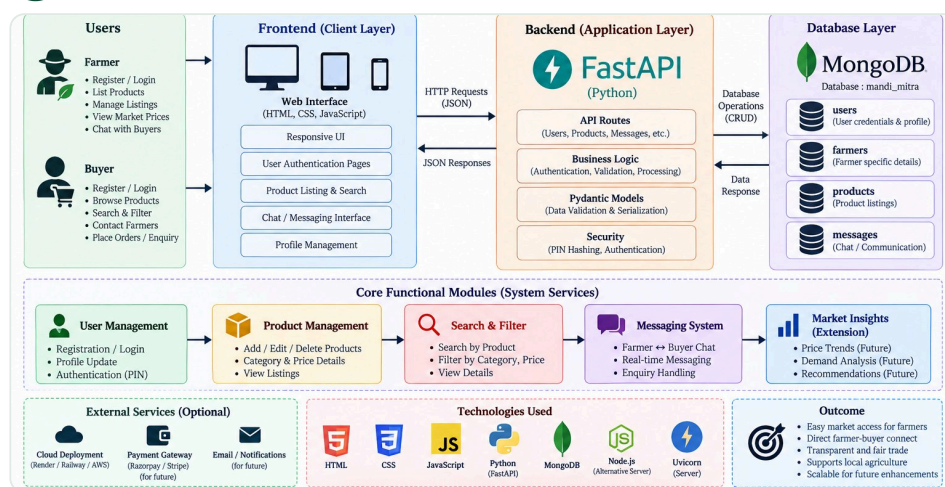
MongoDB

Git & GitHub

Postman

VS Code

S System Architecture

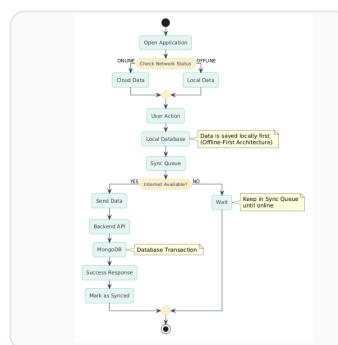


D Database Design (MongoDB)

users _id · phone · pin (hashed) · role	farmers _id · user_id · name · location
products _id · farmer_id · name · price · qty	messages _id · sender · receiver · text
orders · planned _id · buyer · product · status	syncQueue · planned _id · operation · synced

Database **mandi_mitra** → Collections → Documents. users 1:1 farmers · farmers 1:N products · users 1:N messages.

W Offline-First Sync Workflow



- App opens and checks network status.
- Every action is saved to the local database first.
- Changes wait in the sync queue until online.
- Queue is sent via the API to MongoDB and marked as synced.